

Notes: Di Maggio & Kacperczyk (JFE, 2017)

The Unintended Consequences of the Zero Lower Bound Policy

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1 Overview

When short rates are pushed to (and expected to remain at) the zero lower bound (ZLB), the business model of prime money market funds (MMFs) becomes fragile: investors chase tiny yield differences more aggressively, fund sponsors cut fees and/or subsidize returns, and many funds either *reach for yield* (take more credit risk) or *exit*—with spillovers to short-term corporate funding and to the broader mutual-fund industry.

2 Core mechanism

The paper’s logic is a very clean “equilibrium incentive” story:

1. **Mechanical margin compression at ZLB.** Prime MMFs earn short-term yields on a constrained asset menu (Rule 2a-7). When the policy rate collapses toward zero, gross yields collapse almost one-for-one, but operating costs are sticky. So the *net* return to investors is squeezed, sometimes requiring fee waivers/subsidies to avoid negative net yields.
2. **Competitive fragility.** Even when all net yields are near zero, *small* differences in gross yields can still matter a lot for flows. If flows become more sensitive to performance when rates are low, then the payoff to “beating peers by a few bps” increases precisely when fundamentals are worst.
3. **Sponsor heterogeneity matters.** Sponsors with more reputation at stake (e.g., large diversified financial groups) are more willing to exit or to subsidize rather than increase risk. Smaller/independent sponsors have weaker franchise concerns and stronger incentives to reach for yield to keep assets.
4. **Spillovers: corporate funding & industry reallocation.** If prime MMFs contract/exit, issuers that rely on them for commercial paper (CP) and other short-term funding lose an important marginal investor. At the same time, fund families reallocate product supply toward bond funds (a close substitute for yield-seeking investors).

3 Data, sample, and key variable definitions

- **Fund universe:** U.S. taxable *prime* MMFs, weekly panel (roughly 2005–2013 in the main fund-level tests).
- **Holdings:** post-2010 SEC Form N-MFP holdings allow security-level yield/borrower analysis (only available for later ZLB events).
- **Key fund outcomes:**
 - *Fund gross return:* annualized yield (pre-fee).
 - *Flows:* change in total net assets (TNA) net of reinvested returns.

- *Spread*: fund gross return minus a short Treasury benchmark (a “risk premium” notion).
- *Holdings risk*: fraction of the portfolio in relatively riskier instruments (credit-risk channel).
- *Maturity / maturity risk*: weighted-average maturity (interest-rate/liquidity risk channel).
- *Concentration*: Herfindahl-type measure over asset categories (portfolio concentration).
- *Expenses charged vs incurred*: charged is what investors pay (after waivers), incurred is the stated operating cost; *subsidy* is the wedge (incurred minus charged).
- *Exit*: fund closure/liquidation indicator.

4 Empirical strategy and models (all the key equations)

4.1 (1) Policy rate → fund gross returns: pass-through

The first step is to show that MMF gross yields move almost one-for-one with the policy rate:

$$r_{i,t} = \alpha + \beta \text{FedRate}_t + \Gamma' X_{i,t-1} + \varepsilon_{i,t}, \quad (1)$$

where $r_{i,t}$ is the fund’s annualized gross return and $X_{i,t-1}$ are lagged fund controls (size, age, etc.), with variants including fixed effects.

4.2 (2) “Flow fragility”: flows respond to recent performance, especially at low rates

To test whether competition intensifies at ZLB, the paper estimates a flow-performance relationship:

$$\text{Flow}_{i,t} = \alpha + \phi r_{i,t-1} + \Gamma' X_{i,t-1} + \varepsilon_{i,t}, \quad (2)$$

and then allows the sensitivity to differ in low-rate periods by interacting with a low-rate indicator $\mathbb{1}\{\text{FedRate}_t \leq 1\%\}$.

4.3 (3) Forward-guidance “ZLB shocks” as quasi-experiments

At the ZLB, the *current* rate barely moves, so the identification uses forward-guidance announcements that extend the expected duration of near-zero rates (Table 1). The event-study specification is:

$$Y_{i,t} = \alpha + \gamma \text{Event}_t + \Gamma' X_{i,\text{Jan2006}} + \varepsilon_{i,t}, \quad (3)$$

where $Y_{i,t}$ is an outcome (exit/risk/fees), and $X_{i,\text{Jan2006}}$ are *pre-determined* fund controls measured before the ZLB episode (to avoid “bad controls” that respond endogenously to events).

4.4 (4) Sponsor heterogeneity: independent vs affiliated

To test that reputational/franchise considerations matter, the paper estimates:

$$Y_{i,t} = \alpha + \gamma \text{Event}_t + \delta \text{Independent}_i + \theta(\text{Event}_t \times \text{Independent}_i) + \Gamma' X_{i,\text{Jan2006}} + \varepsilon_{i,t}. \quad (4)$$

4.5 (5) Reputation within independent sponsors

Within independent sponsors, reputational capital is proxied by *business risk* (roughly: how much the sponsor has at stake outside MMFs). The interaction model is:

$$Y_{i,t} = \alpha + \gamma \text{Event}_t + \delta \text{BusinessRisk}_i + \theta(\text{Event}_t \times \text{BusinessRisk}_i) + \Gamma' X_{i,\text{Jan2006}} + \varepsilon_{i,t}. \quad (5)$$

4.6 (6) Robustness: continuous rate variation & asymmetric effects near ZLB

Using the full time series of rate changes, the paper estimates:

$$Y_{i,t} = \alpha + \beta \text{FedRate}_t + \eta \text{LowRate}_t + \kappa(\text{FedRate}_t \times \text{LowRate}_t) + \Gamma' X_{i,\text{Jan2006}} + \varepsilon_{i,t}. \quad (6)$$

The interaction κ captures whether the slope of Y with respect to rates is different in the low-rate regime.

4.7 (7) Holdings-based yield shifts after ZLB shocks

With N-MFP holdings, the paper tests whether funds add higher-yielding securities after forward-guidance events:

$$\bar{y}_{i,t} = \alpha + \gamma \text{PostEvent}_t + \Gamma' X_{i,\text{Jan2006}} + \varepsilon_{i,t}, \quad (7)$$

where $\bar{y}_{i,t}$ is the mean yield on newly added holdings (or holdings yields more generally).

4.8 (8) Issuer outcomes around fund closures

To connect fund exit to corporate financing, the paper estimates issuer-level regressions such as:

$$\text{OutDebt}_{j,t} = \alpha + \beta \text{PostClosure}_{j,t} + \delta_j + \gamma_t + \varepsilon_{j,t}, \quad (8)$$

$$\text{Leverage}_{j,t} = \alpha + \beta_1(\text{Post}_t \times \text{Closure}_j) + \beta_2 \text{Closure}_j + \beta_3 \text{Post}_t + \delta_j + \gamma_t + \varepsilon_{j,t}, \quad (9)$$

with issuer fixed effects δ_j and time fixed effects γ_t .

4.9 (9) Industry response: fund-family product reallocation after MMF closures

Finally, at the fund-family level, the paper estimates a difference-in-differences specification:

$$\text{NumFunds}_{f,t}^{(\text{style})} = \alpha + a_1 \text{After}_{f,t} + a_2 \text{Treated}_{f,t} + a_3(\text{After}_{f,t} \times \text{Treated}_{f,t}) + \Gamma' X_{f,t} + \mu_f + \varepsilon_{f,t}, \quad (10)$$

where $\text{style} \in \{\text{money, bond, equity, balanced}\}$ and μ_f are family fixed effects.

5 Results: interpretation of each figure and table

5.1 Table 1: the five forward-guidance “ZLB shocks”

Table 1 lists five key FOMC announcements (Dec 2008; Mar 2009; Aug 2011; Jan 2012; Sep 2012) that either initiated ZLB policy or extended the expected time until lift-off. **Interpretation:** these announcements shift *expected* future short rates without moving the contemporaneous policy rate much (especially post-2008). This is exactly the right setting to isolate a “profitability at ZLB” channel rather than a standard demand/credit channel tied to contemporaneous rate changes.

Table 1

Zero Interest Rate Policy Events (ZIRP).

We report the dates of FOMC meetings in which the Fed decided to change the Fed target rate or provided policy guidance about the prevailing zero interest rate policy.

Date	Event
December 16, 2008	Fed target rate reduced to 0–0.25%
March 18, 2009	Zero rates for “an extended period of time”
August 9, 2011	Zero rates at least until 2013
January 25, 2012	Zero rates at least until 2014
September 13, 2012	Zero rates at least until 2015

Table 2

Fund gross returns and Fed target rate.

The sample is all U.S. prime money market funds over the period January 2005–December 2013. The dependent variable is *Fund gross return* computed as the annualized return. *Fed rate* is the annualized Fed target rate. Control variables include the natural logarithm of fund assets, the natural logarithm of family assets, the expense ratio (charged), fund age, fund flow computed as a percentage change in total net assets from time t to time $t+1$ adjusted for market appreciation, standard deviation of fund flow growth, and an indicator variable equal to one if the fund is offered to institutional investors and zero otherwise. All regressions are at the weekly level. Column 1 includes year-fixed effects, column 2 includes fund-fixed effects, column 3 includes sponsor-fixed effects, and column 4 includes year-fixed and sponsor-fixed effects. Standard errors are clustered at the week level. ***, **, * represent 1%, 5%, and 10% significance, respectively.

	(1)	(2)	(3)	(4)
Fed rate	93.025*** (1.012)	94.370*** (0.963)	62.291*** (5.087)	62.086*** (5.083)
Controls	Yes	Yes	Yes	Yes
Year-fixed effects	No	No	Yes	Yes
Fund-fixed effects	Yes	No	No	No
Sponsor-fixed effects	No	Yes	No	Yes
Observations	98,496	98,496	98,496	98,496

5.2 Table 2: policy rates pass through to MMF gross returns (Eq. 1)

The Fed rate strongly predicts gross MMF returns, with near one-to-one pass-through.

- In columns without time fixed effects, $\hat{\beta} \approx 0.93$ –0.94 (in bps units), i.e., a 100bp increase in the Fed rate raises annualized gross MMF returns by roughly 93–94bp.
- Even with year fixed effects, $\hat{\beta} \approx 0.62$, still large.

Intuition: when the Fed compresses the short end, the fund’s feasible gross return shrinks mechanically. Since expenses do not shrink one-for-one, net yields get squeezed, creating pressure to change behavior (risk/fees/exit).

5.3 Fig. 1: prime MMF assets under management (AUM)

Fig. 1 shows that prime MMF AUM contracts sharply in the ZLB era (post-2011 especially), consistent with the idea that ZLB is a structural stress test for the industry. **I read this as:** not just a crisis-era run (2008), but a persistent regime shift where prime MMFs become less attractive when rates are pinned near zero.

5.4 Table 3: flows chase yields more when rates are low (Eq. 2)

Table 3 establishes a key competitive amplification:

- Flows respond positively to lagged fund returns ($\hat{\phi} > 0$).

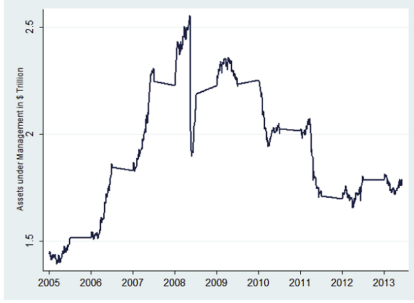


Fig. 1. Assets under management of prime money funds: 2005–2014. Note: The figure presents the evolution of weekly total net assets under management for the universe of U.S. prime money market funds over the period 2005–2014.

Table 3

The flow-performance relationship.

The sample is all U.S. prime money funds over the period January 2005–December 2013. The dependent variable is *Fund flow*, computed as the percentage change in total net assets from time t to time $t+1$, adjusted for market appreciation. *Fed rate* is the annualized Fed target rate. *Fund return* is the annualized fund return. Control variables include the natural logarithm of fund assets, the natural logarithm of family assets, expense ratio (charged), fund age, fund flow computed as a percentage change in total net assets from time t to time $t+1$ adjusted for market appreciation, standard deviation of fund flow, and an indicator variable equal to one if the fund is offered to institutional investors and zero otherwise. All regressions are at the weekly level and include sponsor-fixed effects. *Low rate* restricts the sample to the period of low interest rates (Fed target rate between zero and 1%). Columns 2 and 4 additionally include week-fixed effects. Standard errors are clustered at the fund sponsor level. ***, **, * represent 1%, 5%, and 10% significance, respectively.

	(1)	(2)	(3)	(4)
	Flow		Flow	
Fund return	0.001*** (0.000)	0.009*** (0.001)	0.002*** (0.000)	0.006*** (0.002)
Fund return $\times$ Low rate			0.002** (0.001)	0.006*** (0.002)
Controls	Yes	Yes	Yes	Yes
Sponsor-fixed effects	Yes	Yes	Yes	Yes
Week-fixed effects	No	Yes	No	Yes
Observations	98,948	98,948	98,948	98,948

- The interaction $Fund\ return \times Low\ rate$ is positive and significant, meaning investors chase tiny performance differences *more* when the level of rates is low.

Intuition: at ZLB, everyone earns close to zero, so a few extra basis points become a salient differentiator and can meaningfully shift AUM. That makes yield-seeking (and hence risk-taking) privately valuable.

5.5 Table 4: baseline event-study effects of ZLB shocks (Eq. 3)

This is the core set of results. I read Table 4 as: ZLB shocks trigger *exit*, *reaching for yield*, and *fee compression/subsidies*.

Panel A: exit. After events, the # of funds declines (e.g., about -5.4 within 3 months; -9.0 within 6 months), and fund-level exit probabilities increase (positive γ).

Panel B: risk-taking. After events, funds earn higher *spread* and take more *holdings risk* (credit risk), while *maturity* tends to shorten slightly and *concentration* rises. **Interpretation:** the dominant response is to shift into higher-yield, riskier paper and tilt portfolios (including more concentrated exposures).

Panel C: fees and subsidies. Charged expenses fall (especially at the 6-month horizon), incurred expenses fall somewhat, and subsidies rise (positive at 6 months). **Interpretation:** sponsors partially absorb the margin squeeze to maintain non-negative net yields and avoid outflows.

5.6 Table 5: independent sponsors respond differently (Eq. 4)

Table 5 is the first strong “reputation/franchise” test.

- **Exit:** affiliated sponsors (baseline) are more likely to exit after ZLB shocks; the interaction suggests independent sponsors are *less* likely to exit (especially at 6 months).

Table 4
Fund strategies and ZIRP shocks.
The sample is all U.S. prime MMFs. The estimation window includes one month before and three months or six months after the event dates defined in Table 1. Event is an indicator variable equal to one for the period after the event date and zero for the period before the event date. In Panel A, the dependent variables are Number of funds, defined as the number of funds in a given period, and Exit, defined as an indicator variable equal to one if the fund exits the fund industry in week t . In Panel B, the dependent variables are: the weekly annualized spread (Spread), the fraction of assets held in risky assets, net of the riskless assets (Holdings risk), average portfolio maturity (Maturity risk), and portfolio concentration, defined as a Herfindahl Index of asset classes (Concentration). In Panel C, the dependent variables are Charged expenses, defined as percentage expense rate charged by a fund, Incurred expenses, defined as percentage expense rate incurred by a fund, Subsidy, defined as the difference between incurred and charged expenses. Control variables include the annualized fund return, the natural logarithm of fund assets, the natural logarithm of family assets, expense ratio (charged) fund age, fund flow computed as a percentage change in total net assets from time t to time $t+1$ adjusted for market appreciation, standard deviation of fund flow, and an indicator variable equal to one if the fund is offered to institutional investors and zero otherwise. All regressions are at the weekly level and include year-fixed and sponsor-fixed effects. Standard errors are clustered at the week level. ***, **, * represent 1%, 5%, and 10% significance.

Panel A: Fund exit								
	(1)	(2)	(3)	(4)				
	# Funds	Exit	# Funds	Exit				
	3-Months ahead		6-Months ahead					
Event	-5.400*** (0.543)	0.002*** (0.001)	-9.000*** (1.152)	0.002*** (0.001)				
Controls	No	Yes	No	Yes				
Year-fixed effects	No	Yes	No	Yes				
Sponsor-fixed effects	No	Yes	No	Yes				
Observations	10	18,568	10	25,914				

Panel B: Fund risk								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Spread	Holding risk	Maturity	Concentration	Spread	Holding risk	Maturity	Concentration
	3-Months ahead	3-Months ahead	3-Months ahead	3-Months ahead	6-Months ahead	6-Months ahead	6-Months ahead	6-Months ahead
Event	34.329*** (12.113)	0.954*** (0.323)	-1.216*** (0.503)	0.005*** (0.002)	21.341*** (10.471)	0.879*** (0.271)	-1.347*** (0.386)	0.005*** (0.001)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-month-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sponsor-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	16,830	16,882	16,879	16,882	23,462	23,527	23,524	23,527

Panel C: Fund expenses						
	(1)	(2)	(3)	(4)	(5)	(6)
	Charged	Incurred	Subsidy	Charged	Incurred	Subsidy
	3-Months ahead	3-Months ahead	3-Months ahead	6-Months ahead	6-Months ahead	6-Months ahead
Event	-0.001 (0.002)	-0.002*** (0.001)	-0.002 (0.002)	-0.014** (0.006)	-0.003*** (0.001)	0.010** (0.005)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year-month-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Sponsor-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	18,568	18,506	18,506	25,914	25,829	25,829

Table 5
Fund strategies and ZIRP shocks: Conditioning on sponsor type.
The design follows Table 5. Independent sponsor is an indicator variable equal to one if the fund sponsor is an independent asset management company, and zero otherwise.

Panel A: Fund exit								
	(1)	(2)	(3)	(4)				
	# Funds	Exit	# Funds	Exit				
	3-Months ahead		6-Months ahead					
Event	-3.280*** (0.803)	0.002*** (0.001)	-2.862*** (0.767)	0.002*** (0.001)				
Independent sponsor × Event	0.077 (0.067)	-0.002 (0.002)	0.178** (0.081)	-0.003** (0.001)				
Controls	Yes	Yes	Yes	Yes				
Year-fixed effects	No	Yes	No	Yes				
Sponsor-fixed effects	No	Yes	No	Yes				
Observations	20	18,568	20	25,914				

Panel B: Fund risk								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Spread	Holding risk	Maturity	Concentration	Spread	Holding risk	Maturity	Concentration
	3-Months ahead	3-Months ahead	3-Months ahead	3-Months ahead	6-Months ahead	6-Months ahead	6-Months ahead	6-Months ahead
Independent sponsor	-3.837 (2.878)	6.642** (3.308)	4.768*** (1.366)	0 (0.020)	-0.776 (2.940)	5.681 (3.492)	5.107*** (1.484)	-0.011 (0.021)
Independent sponsor × Event	9.215*** (1.239)	1.821*** (0.634)	-1.010** (0.460)	0.008** (0.004)	4.646*** (1.668)	3.405*** (1.533)	-1.897*** (0.709)	0.026*** (0.007)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	16,830	16,882	16,879	16,882	23,462	23,527	23,524	23,527

Panel C: Fund expenses						
	(1)	(2)	(3)	(4)	(5)	(6)
	Charged	Incurred	Subsidy	Charged	Incurred	Subsidy
	3-Months ahead	3-Months ahead	3-Months ahead	6-Months ahead	6-Months ahead	6-Months ahead
Independent sponsor	-0.003 (0.020)	0.030 (0.039)	0.032 (0.027)	-0.006 (0.022)	0.029 (0.039)	0.034 (0.027)
Independent sponsor × Event	0.005 (0.006)	0.008 (0.005)	0.003 (0.007)	0.011 (0.012)	0.011 (0.010)	0 (0.013)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	18,568	18,506	18,506	25,914	25,829	25,829

- **Risk:** the interaction terms are strongly positive for spread/holdings risk and negative for maturity, implying *independent* sponsors increase risk-taking *more* after events.

Takeaway: who takes risk at ZLB is not random; it is tied to sponsor type. This supports an incentive story (not just “everyone reaches for yield equally”).

5.7 Table 6: within independents, reputation attenuates risk-taking (Eq. 5)

Within independent sponsors, the interaction Event × BusinessRisk is generally:

- **negative** for spread/holdings risk/concentration, and
- **positive** for subsidies.

Interpretation: even among independents, those with more franchise at stake avoid aggressive risk-shifting and instead “pay” (via subsidies) to keep the product viable.

5.8 Fig. 2 and Table 7: nonlinear effects when rates are already low (Eq. 6)

Fig. 2 reminds me the sample contains a long ZLB period. Table 7 shows the response to rate changes is *asymmetric*:

- In the high-rate regime, changes in rates have comparatively modest effects.
- In the low-rate regime, a further decline in rates is associated with:
 - fewer funds / more exits (Panel A),
 - higher spreads and more holdings risk (Panel B), and

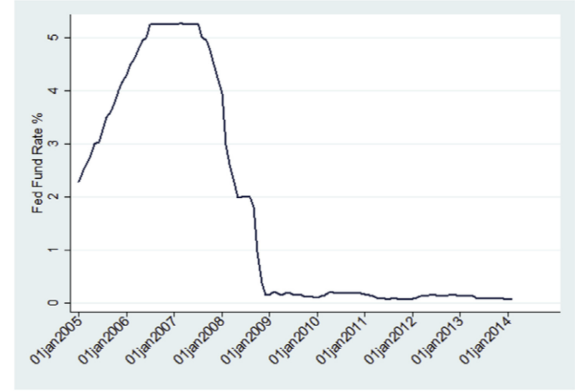
Table 6

Variation in reputation within independent fund sponsors.

The design follows Table 5. *Business risk* is the ratio of assets under management held by a fund family in non-money funds and total assets under management. Panel A presents the results for the 3-month announcement horizon and Panel B for the 6-month horizon.

Panel A: 3-Month horizon						
	Exit	Spread	Holdings risk	Maturity	Concentration	Subsidy
Event	0.001 (0.002)	25.207** (12.275)	4.368* (2.439)	-0.120 (0.936)	0.018* (0.010)	-0.050*** (0.015)
Event× Business risk	-0.001 (0.003)	-11.003* (6.241)	-3.661* (2.349)	-1.774* (1.002)	-0.016 (0.012)	0.062*** (0.019)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sponsor F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Year F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7635	7030	7036	7036	7036	7619

Panel B: 6-Month horizon						
	Exit	Spread	Holdings risk	Maturity	Concentration	Subsidy
Event	-0.001 (0.004)	22.596** (10.835)	7.848*** (2.016)	2.395** (0.958)	0.036*** (0.008)	-0.052*** (0.011)
Event× Business risk	-0.001 (0.005)	-11.003** (4.826)	-8.226*** (2.367)	-4.705*** (1.113)	-0.039*** (0.009)	0.079*** (0.015)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sponsor F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Year F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10,659	9789	9802	9802	9802	10,633

**Fig. 2.** Fed target rate: 2005–2014.

- more fee pressure and larger subsidies (Panel C).

Intuition: the ZLB is not “just another low-rate point”; it is a region where fixed costs and institutional constraints make behavior kinked/nonlinear.

Table 7

Comparisons across interest rate regimes.

The sample is all U.S. prime money market funds over the period January 2005–December 2013. The definitions of all dependent and control variables follow those in Table 5. All regressions are at the weekly level and include year/month-fixed and sponsor-fixed effects. *Low rate* is an indicator variable equal to one if Fed target rate is below 1%, and zero otherwise. Standard errors are clustered at the week level. ***, **, * represent 1%, 5%, and 10% significance, respectively.

Panel A: Fund exit				
	(1)	(2)	(3)	(4)
	# Funds		Exit RF > 1	
Fed rate	-1.555*** (0.475)	-2.348*** (0.258)	0.003 (0.002)	-0.072*** (0.016)
Low rate		-132.783*** (12.459)		
Fed rate×Low rate		16.861*** (3.171)		
Controls	Yes	Yes	Yes	Yes
Year/Month-F. E.	No	No	Yes	Yes
Sponsor F. E.	No	No	Yes	Yes
Observations	442	442	50,334	48,458

Panel B: Fund risk				
	(1)	(2)	(3)	(4)
	Spread	Holdings risk	Maturity risk	Concentration
Fed rate	2.713 (26.732)	-0.048 (0.28)	-0.893 (0.649)	0.001 (0.001)
Fed rate×Low rate	-56.611*** (13.628)	-5.941*** (0.723)	-1.673* (1.032)	-0.028*** (0.006)
Controls	Yes	Yes	Yes	Yes
Sponsor-fixed effects	Yes	Yes	Yes	Yes
Year/Month-fixed effects	Yes	Yes	Yes	Yes
Observations	94,521	95,264	95,253	94,264

Panel C: Fund expenses						
	(1)	(2)	(3)	(4)	(5)	(6)
	Charged expenses		Incurred expenses		Subsidy	
Fed rate	-0.006*** (0.001)	-0.005*** (0.001)	-0.002 (0.002)	0.004*** (0.001)	0.004*** (0.001)	0.008*** (0.001)
Fed rate×Low rate		0.080*** (0.027)		-0.001 (0.011)		-0.079*** (0.022)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sponsor-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	98,484	98,484	98,484	98,484	98,484	98,484

Table 8

The effect of longevity risk.

The sample is all U.S. prime MMFs over the period January 2005–December 2013. The definitions of all dependent variables follow those in Table 5. *Longevity* is defined as the natural logarithm of 1 + the number of days that would take for the interest rate (30-day) to exceed 1%. The values are interpolated off the future yield curve. All regressions are at the weekly level and include year/month-fixed and sponsor-fixed effects. We focus on the case in which the Fed target rate is below 1%. Standard errors are clustered at the week level. ***, **, * represent 1%, 5%, and 10% significance, respectively.

	Exit	Spread	Holdings risk	Maturity	Concentration	Subsidy
Longevity	0.0012 (0.0008)	29.937*** (7.046)	0.0090*** (0.0017)	-0.193 (0.193)	0.00300*** (0.00120)	0.030*** (0.006)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Sponsor F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Year F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Observations	48,458	45,322	46,286	46,054	46,065	48,298

5.9 Table 8: expected duration (longevity) strengthens incentives

Table 8 uses a measure of expected time until rates rise meaningfully. Longer expected ZLB duration predicts:

- higher spreads and holdings risk,
- higher concentration,
- larger subsidies.

Interpretation: forward guidance matters because it changes the *horizon* over which the margin squeeze persists, strengthening incentives to search for yield or to subsidize.

5.10 Fig. 3 and Table 9: security-level evidence + corporate spillovers

Fig. 3. The density plots show yields on newly added holdings shift to the right after the 2011–2012 guidance shocks (vs placebo dates).

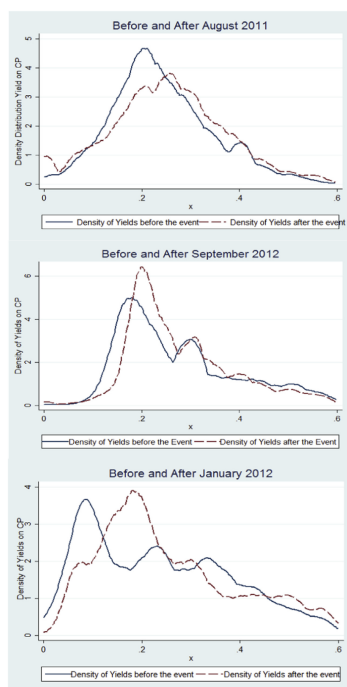


Fig. 3. Yields distribution of CP (Commercial Paper) and ZIRP shocks. Note: Empirical distributions of the fund portfolio yields before and after the three ZIRP shocks.

Table 9

Evidence from portfolio holdings.

The sample is all U.S. prime MMFs. The dependent variable is the mean of the yields of the new securities added to the fund portfolio. The estimation window includes one month before and three months after the last three event dates defined in Table 1. Event is an indicator variable equal to one for the period after the event date and zero for the period before the event date. Columns 4 and 5 repeat the analysis by considering a randomly drawn date that does not coincide with any of the ZIRP shocks dates. All regressions are at the monthly level and include year/month-fixed and fund-fixed effects. Standard errors are clustered at the monthly level. ***, **, * represent 1%, 5%, and 10% significance, respectively.

Panel A		(1)	(2)	(3)	(4)	(5)
		Mean yield				
		Event: August 2011	Event: January 2012	Event: September 2012	March 2011 January 2013	Placebos
Event		0.0388*** (0.000701)	0.0439*** (0.000971)	0.0420*** (0.000371)	-0.0302*** (0.000438)	-0.0107*** (0.000287)
Controls		Yes	Yes	Yes	Yes	Yes
Year/month-fixed effects		Yes	Yes	Yes	Yes	Yes
Fund-fixed effects		Yes	Yes	Yes	Yes	Yes
Observations		49,242	39,690	59,716	31,988	24,129

Panel B		(1)	(2)
Number of borrowers per fund			
Event		-0.423** (0.209)	-0.378** (0.182)
Fund-fixed effects		Yes	Yes
Time trend		No	Yes
Observations		1,288	1,288
R-squared		0.902	0.902

Panel C		(1)	(2)
		Outstanding debt	
Post fund closure		-0.419*** (0.0686)	-0.153** (0.0605)
Issuer-fixed effects		Yes	Yes
Month-fixed effects		No	Yes
Observations		52,682	52,682
R-squared		0.492	0.524

Panel D		(1)	(2)	(3)	(4)
		Leverage 6-month window	Leverage 6-month window	Avg. leverage 6-month window	Leverage 1-year window
Post × Closure		-0.006** (0.003)	-0.006* (0.003)	-0.020* (0.011)	-0.004 (0.008)
Post		0.001 (0.002)	-0.000 (0.001)	0.007 (0.004)	0.002 (0.007)
Closure		0.004 (0.012)	0.013** (0.005)	0.016* (0.008)	0.003 (0.004)
Time-fixed effects		Yes	Yes	Yes	Yes
Issuer-fixed effects		No	Yes	Yes	Yes
Observations		895	895	448	1,784
R-squared		0.005	0.918	0.929	0.909

Table 9, Panel A. Regressions confirm a statistically sharp increase in yields on newly added securities right after the three post-2010 events; placebo dates do not show such increases.

Table 9, Panel B. The # of borrowers per fund declines post-event: the reach-for-yield response is accompanied by more concentrated lending relationships.

Table 9, Panels C–D. Fund closures reduce issuers’ outstanding short-term debt and are associated with declines in issuer leverage (diff-in-diff). **Takeaway:** this is not only a “portfolio choice” story—it transmits to the real financing environment for firms that rely on MMFs.

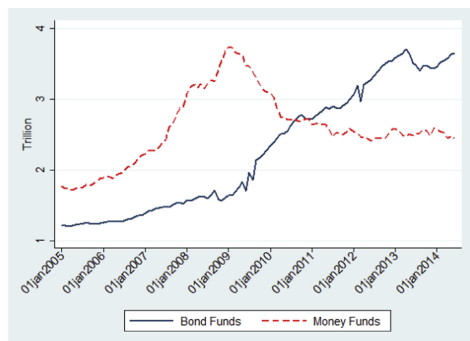


Fig. 4. Money funds vs. bond funds AUM: 2005–2014. The figure presents the evolution of weekly total net assets for the universe of U.S. prime money market funds and bond funds over the period 2005–2014.

Table 10
Industry response.
The sample includes all mutual funds families in U.S. over the period January 2005–December 2013. The dependent variable is the number of funds (at share class level) in a given period for a given type of fund within a fund family. Panel A groups funds into money, bond, equity, and balanced. Panel B divides bond funds into short-term, medium-term, and long-term. LR denotes low interest rate regime and HR denotes high interest rate regime. Treated is an indicator variable equal to one if the fund family closed one of its MMFs and zero for all other funds. After is an indicator variable equal to one six months after the closure of the MMF and zero six months before the fund closure. Control variables are family return and natural logarithm of family size. All regressions include fund family-fixed effects. Standard errors are clustered at the year-month level. “”, “”, “” represent 1%, 5%, and 10% significance, respectively.

Panel A: Asset allocations							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Money funds		Bond funds		Equity funds		Balanced funds
	LR	HR	LR	HR	LR	HR	LR
After	-0.113 (0.068)	0.000 (0.500)	0.070 (0.206)	-0.068 (0.198)	0.420 (0.654)	-0.275 (0.546)	0.174 (0.158)
After×Treated	-3.313** (1.528)	-0.088 (1.657)	7.098** (3.620)	-1.032 (1.987)	16.932 (13.802)	0.818 (3.455)	3.944* (2.255)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Family-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,951	2,834	2,951	2,834	2,951	2,834	2,951

Panel B: Bond maturity					
	(1)	(2)	(3)	(4)	(5)
	Short-term bonds		Medium-term bonds		Long-term bonds
	LR	HR	LR	HR	HR
After	0.105 (0.067)	0.090** (0.041)	0.765* (0.391)	-4.410 (3.095)	-0.799** (0.316)
After×Treated	0.757** (0.336)	0.397 (0.479)	1.694 (1.103)	-3.678 (9.684)	4.738* (2.669)
Controls	Yes	Yes	Yes	Yes	Yes
Family-fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	2,951	2,834	2,951	2,834	2,951

5.11 Fig. 4 and Table 10: industry reallocation after MMF closures

Fig. 4 shows money-fund AUM declines while bond-fund AUM rises post-crisis, suggesting substitution. Table 10 makes this sharp using family-level diff-in-diff:

- In the low-rate regime, families that close MMFs reduce the number of money funds and increase the number of bond funds (and somewhat balanced funds).
- There is little evidence of comparable reallocation in high-rate regimes.

Interpretation: ZLB pushes families to redesign their product menu, likely because investors and fee revenue migrate to “near substitutes” (bond funds).

5.12 Table 11: what happens to managers after MMF closures

Table 11 is descriptive but helpful: many managers remain in the same family (often continuing to manage MMFs or moving to other products), while a nontrivial share transitions across families or out of the sector. **Interpretation:** closures look like a strategic firm decision and an industry reallocation, not just manager failure.

Table 11
The effect of MMF closure on managers’ rotation.
The sample is conditional on the fund closing its operations. We track managers’ subsequent career outcomes. Possible career outcomes are: (1) staying in the same family and running a MMF; (2) staying in the same family and running a non-money market fund; (3) going to a different family and running a MMF; (4) going to a different family and running a non-money market fund; (5) leaving the fund industry (moving to private business, government institutions, setting up own company, etc.). We condition on a single manager departure in a given date (a particular multiple fund departure is called as one case).

Career outcome	MMF in the same family	Non-MMF in the same family	MMF in another family	Non-MMF in another family	Other finance career
# of Cases	61	30	35	22	40
% of Cases	32.4%	16.0%	18.6%	11.7%	21.3%

6 Short summary

1. **A clean identification template at ZLB:** use forward-guidance shocks (path surprises) to study behavior when the level rate is stuck.
2. **Endogenous risk-taking under binding constraints:** the paper shows how a policy regime can reshape incentives and induce risk-shifting/exit.
3. **Spillover measurement:** linking intermediary-side responses (fund behavior) to borrower outcomes and to industry reallocation.

7 Conclusion

7.1 Summary: what the paper does

This paper studies how the **zero lower bound (ZLB)** and **forward guidance** (i.e., policy communication about how long rates will stay near zero) affect the **industrial organization** of the U.S. money market fund (MMF) industry. The key empirical idea is that once the policy rate is at (or near) zero, the *duration* of the low-rate regime becomes the economically relevant shock for MMFs, because MMFs hold short-term instruments tied to the Fed target rate and earn very little spread above zero.

Using narrow windows around major FOMC ZLB/forward-guidance announcements, the paper documents that sustained low rates change (i) **portfolio choices** (risk), (ii) **pricing/fees** (expense waivers), and (iii) **entry/exit dynamics** (fund closures), with economically meaningful **spillovers to short-term corporate credit supply**.

7.2 Main conclusions

1. **Reaching for yield / risk-taking:** during unusually low interest rates, MMFs *increase portfolio risk* (within the constraints of Rule 2a-7) to deliver nonnegative net yields to investors.
2. **Large implicit subsidies via fee cuts/waivers:** to avoid negative net returns, MMFs substantially *reduce the expenses charged to investors* even though incurred operating costs do not fall proportionally. The implied sponsor/investor subsidy is economically large: about **\$27 million per average fund** and roughly **\$7.3 billion in aggregate**.
3. **Higher exit/closure probability:** funds that cannot retain their investor base (or fear reputational spillovers) are *more likely to exit*. Thus, ZLB policy reshapes the *market structure* of the MMF sector via consolidation and closures.
4. **Real and market-wide consequences:** the ZLB period is associated with a *reduction in capital supply* to sectors relying on MMF funding (especially large corporate and financial issuers). Empirically, firms borrowing from MMFs that close experience a decline in outstanding MMF-provided short-term debt on the order of **15% or more** (interpretable as deleveraging / reduced access to this funding source).

5. **Higher fragility (runs/default exposure):** the combination of (i) higher risk-taking and (ii) strained profitability increases the system’s exposure to *costly runs and default-type events*. A complementary indicator is that, during the low-rate regime, many funds file *letters of support* with regulators, consistent with elevated expected default pressure.
6. **Industry reallocation beyond MMFs:** closures do not only destroy activity; they *reallocate* it. Fund families that close MMFs subsequently expand other products, especially bond funds. In the low-rate regime, treated families (those experiencing MMF closures) open **many more bond funds** (on the order of **six additional bond funds**) relative to control families. Manager careers also shift away from MMFs: roughly **half** of departing managers do *not* return to managing MMFs.

7.3 Economic intuition: why ZLB changes the MMF equilibrium

The paper’s logic can be summarized as a tight **profitability–fragility tradeoff** created by ZLB:

1. **MMFs are “short-rate” intermediaries with a stable-value norm.** MMFs invest in short-term instruments whose yields move closely with the Fed target rate, while many investors treat MMFs as a cash-like product.
2. **When rates approach zero, the business model becomes mechanically stressed.** Gross yields compress toward zero, but operating costs do not vanish. If expenses were passed through fully, net yields could become near-zero or negative, threatening investor demand and the stable-value franchise.
3. **Funds therefore adjust on three margins:**
 - *Risk margin (reach for yield):* tilt portfolios toward higher-yielding (but riskier) eligible assets.
 - *Price margin (fee waivers):* reduce/waive fees to keep net yields nonnegative, effectively subsidizing investors.
 - *Selection/exit margin:* funds that cannot compete (or whose sponsors will not subsidize) close or merge.
4. **These private responses generate aggregate externalities.** Risk-taking can raise run fragility; exit shrinks an important segment of short-term credit supply; and resource reallocation pushes assets and talent into other fund sectors (e.g., bond funds), changing the allocation of capital in the broader asset management industry.

7.4 Interpretation and scope (what the paper *does not* claim)

The paper emphasizes that its findings suggest economically adverse adjustments, but a **full welfare analysis** is required to sign the net effect. It also stresses that the **reaching-for-yield channel is not unique to MMFs**: other intermediaries (e.g., insurers and pension funds) may react to prolonged low rates by shifting toward riskier asset exposures. Hence, the results are

best read as evidence that **ZLB policy can reshape financial intermediation and market structure**, not only move asset prices.

7.5 Takeaway

At the ZLB, forward guidance about “rates staying low for longer” compresses MMF profitability and triggers a three-margin response—risk-taking, fee subsidies, and exit—which reshapes the MMF industry and reduces short-term credit supply, raising the system’s exposure to runs and default-type stress.